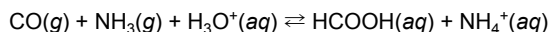
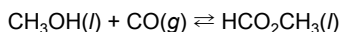


Formic acid (HCOOH), the simplest carboxylic acid, occurs naturally in a variety of sources including the venom of bee and ant stings. Due to its commercial availability, HCOOH is commonly used as a preservative and antibacterial agent in the food and pharmaceutical industries. Various industrial methods for synthesizing HCOOH have emerged in response to an increased demand. For example, one synthetic route produces HCOOH through the spontaneous reaction between gaseous CO and NH₃ in the presence of an acidified aqueous medium.



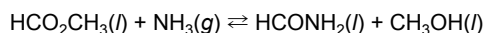
Reaction 1

Formation of HCOOH through Reaction 1 occurs via a three-step reaction mechanism. The first step proceeds according to Reaction 2 and is catalyzed by sodium methoxide.

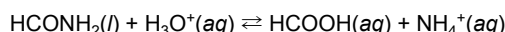


Reaction 2 (slow step)

The second and third steps of the mechanism proceed according to Reactions 3 and 4, respectively.



Reaction 3 (fast step)



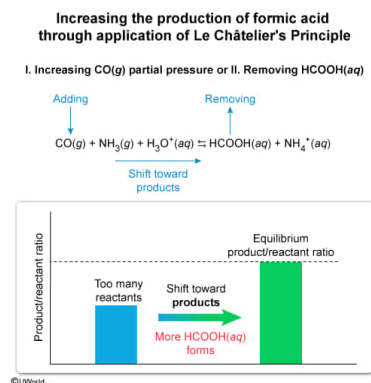
Reaction 4 (fast step)

The amount of formic acid produced from Reaction 1 can be increased by doing which of the following?

- I. Increasing the partial pressure of CO(g)
 - II. Continuously removing HCOOH(aq) from the reaction mixture
 - III. Decreasing the partial pressure of NH₃(g)
- ☐ A. II only [8%]
☐ B. III only [4%]
☒ C. I and II only [80%]
☐ D. I, II, and III [6%]

Correct
80% answered correctly

Explanation:



Le Châtelier's principle says that if the **equilibrium state** of a system is perturbed (eg, **removal or addition** of a reactant or a product), the system will shift toward the reactants or toward the products to counteract the disturbance and establish a new equilibrium. **Removing products or adding reactants** causes the reaction to **form more products** (ie, the equilibrium shifts to the right) whereas adding products or removing reactants causes the reaction to form more reactants (ie, the equilibrium shifts to the left).

The question asks which change to the system would result in an increase in the amount of HCOOH(aq) produced in Reaction 1. Increasing the partial pressure of CO(g) (ie, adding a reactant) would cause the equilibrium to shift toward the products, which would increase the amount of HCOOH(aq) formed (**Number I**). In addition, continuously removing HCOOH(aq) from the reaction mixture as it is formed would also cause the equilibrium to shift toward the products, resulting in more HCOOH(aq) produced (**Number II**).

(**Number III**) **Decreasing** the partial pressure of NH₃(g) (ie, removing a reactant) would cause the equilibrium to shift to the left, which

would form more reactants and cause $\text{HCOOH}(aq)$ to be consumed.

Educational objective:

Le Châtelier's principle states that if a reaction mixture at equilibrium undergoes an alteration in either reactant or product concentration, the system will shift in the direction that opposes the change. Removing products or adding reactants causes the equilibrium to shift to the right whereas adding products or removing reactants causes the equilibrium to shift to the left.

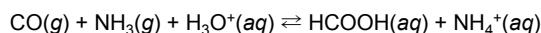
Time spent: 0 sec

Subject:
General Chemistry

QID: 403526

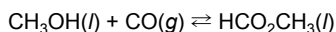
System:
Thermodynamics, Kinetics & Gas Laws

Formic acid (HCOOH), the simplest carboxylic acid, occurs naturally in a variety of sources including the venom of bee and ant stings. Due to its commercial availability, HCOOH is commonly used as a preservative and antibacterial agent in the food and pharmaceutical industries. Various industrial methods for synthesizing HCOOH have emerged in response to an increased demand. For example, one synthetic route produces HCOOH through the spontaneous reaction between gaseous CO and NH₃ in the presence of an acidified aqueous medium.



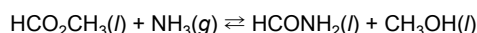
Reaction 1

Formation of HCOOH through Reaction 1 occurs via a three-step reaction mechanism. The first step proceeds according to Reaction 2 and is catalyzed by sodium methoxide.

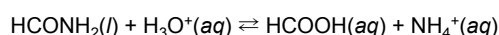


Reaction 2 (slow step)

The second and third steps of the mechanism proceed according to Reactions 3 and 4, respectively.



Reaction 3 (fast step)



Reaction 4 (fast step)

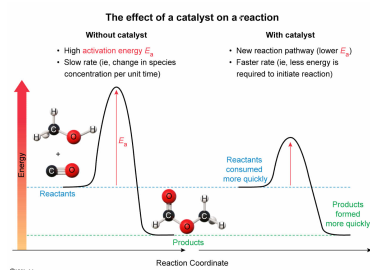
The role of sodium methoxide in Reaction 2 is to:

- ✔ ☒ A. increase the amount of CH₃OH(l) and CO(g) consumed per unit of time. [73%]
- ☐ B. increase the total amount of HCO₂CH₃(l) produced. [9%]
- ☐ C. decrease the number of productive collisions between CH₃OH(l) and CO(g). [2%]
- ☐ D. decrease the rate of the reverse reaction. [14%]

Correct

73% answered correctly

Explanation:



The **rate of a reaction** indicates how quickly the reactant molecules are consumed and the product molecules produced (ie, the change in the species concentrations per unit time). A **catalyst** is a substance that **increases the reaction rate** but is **not consumed** in the overall reaction. Catalysts work by providing an alternative reaction mechanism that forms a more stable (ie, **less energetic**) **transition state**, and thereby **lowers** the **activation energy** E_a of the **rate-determining step** (ie, less energy is required to initiate the reaction).

The passage states that Reaction 2 is catalyzed by sodium methoxide. Therefore, sodium methoxide lowers the E_a of the reaction, which increases the amount of CH₃OH(l) and CO(g) consumed per unit of time (ie, the rate of reactant consumption increases).

(Choice B) The sodium methoxide catalyst increases the amount of HCO₂CH₃(l) produced *per unit time* (ie, how *fast* the product forms), not the total *amount* of HCO₂CH₃(l) produced.

(Choice C) A **catalyst increases** the number of **productive collisions** between reactant molecules that result in a reaction.

(Choice D) The rates of both the forward and reverse reactions *increase* in the presence of the sodium methoxide catalyst.

Educational objective:

Catalysts decrease the activation energy of the rate-determining step of a reaction, thereby increasing the reaction rate (ie, the change in species concentration per unit of time).

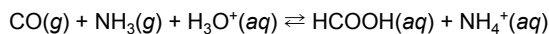
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Subject:
General Chemistry

QID:403527

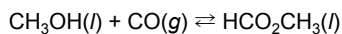
System:
Thermodynamics, Kinetics & Gas Laws

Formic acid (HCOOH), the simplest carboxylic acid, occurs naturally in a variety of sources including the venom of bee and ant stings. Due to its commercial availability, HCOOH is commonly used as a preservative and antibacterial agent in the food and pharmaceutical industries. Various industrial methods for synthesizing HCOOH have emerged in response to an increased demand. For example, one synthetic route produces HCOOH through the spontaneous reaction between gaseous CO and NH₃ in the presence of an acidified aqueous medium.



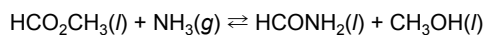
Reaction 1

Formation of HCOOH through Reaction 1 occurs via a three-step reaction mechanism. The first step proceeds according to Reaction 2 and is catalyzed by sodium methoxide.

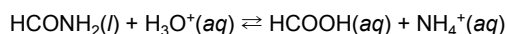


Reaction 2 (slow step)

The second and third steps of the mechanism proceed according to Reactions 3 and 4, respectively.



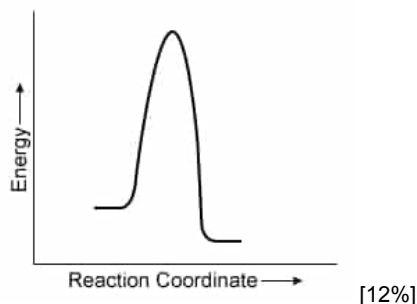
Reaction 3 (fast step)



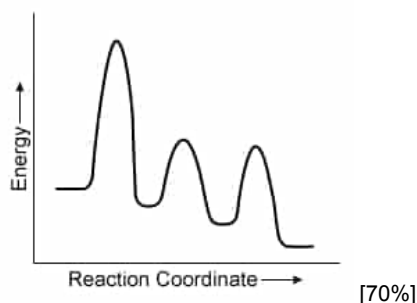
Reaction 4 (fast step)

Which of the following diagrams best illustrates the energy profile for Reaction 1?

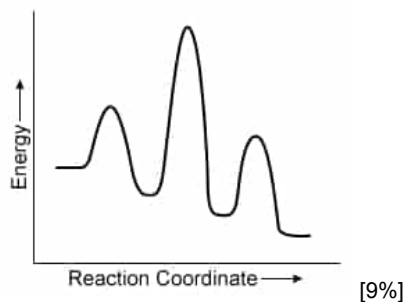
☐ A.



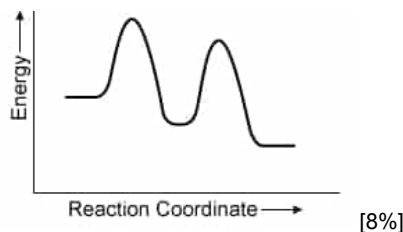
✓ ☒ B.



☐ C.

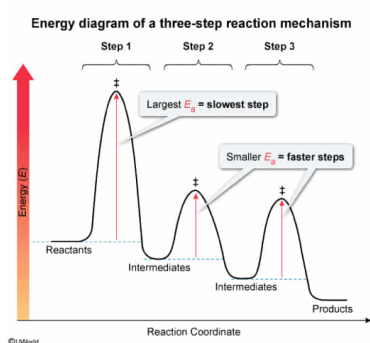


☐ D.



Correct
70% answered correctly

Explanation:



The energy change that occurs over the duration of a reaction is depicted by a **reaction energy diagram**, which is a plot of energy (y-axis) versus reaction progress as represented by the **reaction coordinate** (x-axis). The reaction coordinate reflects the change in molecular geometry of the system, starting with the reactants and ending with the products.

During an **elementary reaction**, existing bonds are broken and new bonds are formed, resulting in a momentary **transition state** located at an **energy maximum** (ie, a peak on the energy diagram). As such, multistep reactions have a corresponding transition state for each step (ie, one peak per step), and the **intermediates** formed during the multistep process appear as "valleys" between the peaks.

The energy difference between the reactants (or intermediates) and the transition state (ie, the **activation energy** E_a) affects the reaction rate. An elementary reaction with a **large** E_a will have a **small rate constant** (ie, a "slow" step) because **more energy** is required to initiate the reaction. Conversely, an elementary reaction with a **small** E_a will have a **large rate constant** (ie, a "fast" step) because **less energy** is required to initiate the reaction.

According to the passage, the formation of formic acid occurs in three steps, and step 1 has a slower rate than steps 2 and 3 (fast steps). This indicates that the reaction energy diagram should have three peaks, and the E_a peak for step 1 should be larger than the E_a peaks for steps 2 and 3. Therefore, the energy diagram shown in Choice B best matches the three-step reaction mechanism.

(Choice A) This energy diagram corresponds to a *one-step* reaction mechanism (ie, one peak).

(Choice C) Because step 1 is *slow* in the three-step mechanism, the E_a for step 1 is greater than the E_a for step 2 and step 3.

(Choice D) An energy diagram with two peaks corresponds to a *two-step* reaction mechanism, but the passage states that Reaction 1 follows a three-step reaction mechanism.

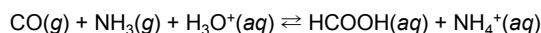
Educational objective:

In a reaction energy diagram, one transition state peak is seen for each elementary step in the reaction mechanism. The greater the difference in energy between the reactants and the transition state, the slower the reaction rate and vice versa.

Time spent: 0 sec
Subject:
General Chemistry

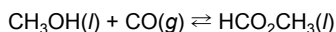
QID: 403528
System:
Thermodynamics, Kinetics & Gas Laws

Formic acid (HCOOH), the simplest carboxylic acid, occurs naturally in a variety of sources including the venom of bee and ant stings. Due to its commercial availability, HCOOH is commonly used as a preservative and antibacterial agent in the food and pharmaceutical industries. Various industrial methods for synthesizing HCOOH have emerged in response to an increased demand. For example, one synthetic route produces HCOOH through the spontaneous reaction between gaseous CO and NH₃ in the presence of an acidified aqueous medium.



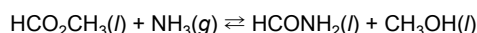
Reaction 1

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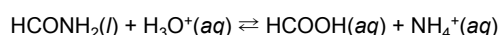


Reaction 2 (slow step)

The second and third steps of the mechanism proceed according to Reactions 3 and 4, respectively.



Reaction 3 (fast step)



Reaction 4 (fast step)

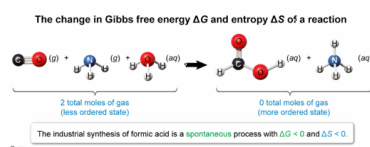
Which of the following correctly describes the total change in Gibbs free energy and the change in entropy, respectively, during Reaction 1?

- ☐ A. $\Delta G > 0$, $\Delta S < 0$ [8%]
☐ B. $\Delta G < 0$, $\Delta S > 0$ [40%]
☐ C. $\Delta G > 0$, $\Delta S > 0$ [2%]
☒ D. $\Delta G < 0$, $\Delta S < 0$ [48%]

Correct

49% answered correctly

Explanation:



The change in **Gibbs free energy** ΔG of a chemical reaction indicates the **spontaneity** (ie, thermodynamic favorability) of the process. A value of $\Delta G < 0$ (ie, a negative value) indicates that the reaction is **spontaneous** (ie, favorable), and a value of $\Delta G > 0$ (ie, a positive value) indicates that the reaction is **nonspontaneous** (ie, unfavorable).

In the passage, the industrial synthesis of formic acid is stated as being a spontaneous reaction. Consequently, $\Delta G < 0$ for the overall reaction.

Entropy S is a measure of the molecular disorder of a system. **Solids** contain **atoms in a fixed arrangement** with little room to move, resulting in **relatively low entropy**. Conversely, **gases** have **relatively high entropy** because they are **free to move** and reconfigure into many different arrangements. The entropy of liquids is between that of solids and gases ($S_{\text{solid}} < S_{\text{liquid}} < S_{\text{gas}}$).

During a reaction, the entropy of a system changes and the change in entropy is calculated as $\Delta S = S_{\text{final}} - S_{\text{initial}}$. The process of converting a gas to a liquid or a solid generally involves a **decrease** in ΔS . According to Reaction 1, the 2 moles of gas and 1 mole of aqueous species react to form 2 moles of an aqueous species. Because the initial state of the system is more disordered (ie, 2 moles of gas) than the final state (ie, only aqueous species and 0 moles of gas), $\Delta S < 0$ for the reaction (ie, entropy decreases, and ΔS is negative).

(Choices A, B, and C) Reaction 1 occurs spontaneously (ie, $\Delta G < 0$) and produces a more ordered final state (ie, $\Delta S < 0$).

Educational objective:

The change in Gibbs free energy ΔG indicates whether a reaction is spontaneous ($\Delta G < 0$) or nonspontaneous ($\Delta G > 0$). Entropy describes the disorder of a molecular system. Gases are disordered (ie, high entropy) and solids are highly ordered (ie, low entropy). The entropy of liquids is between that of solids and gases.

Time spent: 0 sec
 Subject:
 General Chemistry

QID: 403529
 System:
 Thermodynamics, Kinetics & Gas Laws